

```
import pandas as pd, numpy as np  
import seaborn as sns, matplotlib.pyplot as plt
```

```
d = pd.DataFrame({"X":range(1,11),  
"Y":[2.3,2.9,3.8,4.2,5.1,5.5,6.8,7.2,8]})
```

```
print(d)
```

```
sns.regplot(x="X",y="Y",data=d); plt.show()
```

```
sns.histplot(d["Y"],kde=True); plt.show()
```

```
sns.pairplot(d); plt.show()
```

```
A=np.array([[2,3],[5,7]])
```

```
b=np.array([8,19])
```

```
print("Solution:",np.linalg.solve(A,b))
```

```
print("Det:",np.linalg.det(A))
```

```
print("Rank:",np.linalg.matrix_rank(A))
```

```
print("Eigen:",np.linalg.eigvals(A))
```